ContactScale: resolving thermal contact before diffusion hides it

Computational study, 19 October 2026

Abstract

Can an accurate late-time heat flux conceal an unresolved initial contact? This reproducible study compares a conservative finite-volume chain with a semi-infinite analytical solution for two solids separated by a constant thermal contact resistance. Modal propagation removes time-stepping error from the comparison. For independently chosen synthetic properties, flux convergence at one second approaches order 2.0049, yet even the finest tested mesh retains a substantial initial flux deficit. The difference follows directly from the half-cell thermal resistances. Analytical limits, an independent matrix exponential, a surface-impedance integral and separate domain and smooth-solution checks distinguish algebraic, spatial and model errors. These are verification experiments, not physical validation or a claim of a new heat-transfer law.

1 Research question and continuum model

Contact resistance introduces an early response scale that a mesh can fail to resolve even when a later temperature or flux appears converged. The question is how to expose this mismatch without confusing it with time-integration error. The study is motivated by thermal resistance, effusivity and interface reconstruction in DTU MMME lectures 2–4, inspected read-only. All code, text, parameters and figures here are original; no course data or solutions are used.

Consider one-dimensional conduction on each side of a stationary interface:

$$c_i \partial_t T_i = k_i \partial_{xx} T_i, \qquad q = -k_L T_{L,x}(0^-, t) = -k_R T_{R,x}(0^+, t), \qquad T_L(0^-, t) - T_R(0^+, t) = Rq. \quad (1)$$

Here c_i is volumetric heat capacity, k_i conductivity and R area-specific contact resistance. Initially each half-space has a uniform temperature, with gap $\Delta = T_{L,0} - T_{R,0} > 0$. Properties and R are constant; there is no interfacial storage, radiation, phase change, convection or mechanical coupling.

Set $e_i = \sqrt{k_i c_i}$ and $b = e_L^{-1} + e_R^{-1}$. The half-space surface impedances give, in the Laplace variable s,

$$\widehat{q}(s) = \frac{\Delta}{Rs + b\sqrt{s}}, \qquad q(t) = \frac{\Delta}{R}\,\mathrm{erfcx}(z), \quad z = \frac{b\sqrt{t}}{R}, \quad t_c = (R/b)^2. \quad (2)$$

The implementation uses the scaled complementary error function directly [2]. Its large-argument expansion [1] gives $q \sim \Delta/(b\sqrt{\pi t})$ for $t \gg t_c$, while $q(0) = \Delta/R$ for $R > 0$. Perfect contact instead has a singular initial flux: its limit is not uniform near $t = 0$.

2 Numerical method and implementation

Integrating Eq. (2) yields the transferred heat per unit area,

$$Q(t) = \frac{\Delta\sqrt{t}}{b}\left[\frac{2}{\sqrt{\pi}} + \frac{\operatorname{erfcx}(z) - 1}{z}\right]. \tag{3}$$

For $z < 0.1$, direct subtraction is replaced by $Q = (\Delta t/R)\sum_{n=0}^{19} 2a_n z^n/(n+2)$, where $a_0 = 1$, $a_1 = -2/\sqrt{\pi}$ and $a_n = 2a_{n-2}/n$. The perfect-contact branch is $Q = 2\Delta\sqrt{t/\pi}/b$. An independent quadrature verifies the integral and the series transition.

For cell widths h_j, the finite-volume capacities and face conductances are

$$M_{jj} = c_j h_j, \qquad g_j = \left(\frac{h_j}{2k_j} + R_j + \frac{h_{j+1}}{2k_{j+1}}\right)^{-1}, \qquad M\dot{\boldsymbol{T}} = -K\boldsymbol{T}, \quad K = B^T \operatorname{diag}(g) B. \tag{4}$$

The difference operator B produces adjacent temperature differences; only the material interface has nonzero R_j. Outer faces are insulated. In exact arithmetic this connected chain preserves $\mathbf{1}^T M\boldsymbol{T}$ and dissipates $\frac{1}{2}\boldsymbol{T}^T M\boldsymbol{T}$, since $\boldsymbol{T}^T K\boldsymbol{T} = \sum_j g_j(T_j - T_{j+1})^2 \geq 0$.

The Python implementation diagonalizes the symmetric tridiagonal matrix $A = M^{-1/2}KM^{-1/2} = V\Lambda V^T$ using SciPy's LAPACK-backed routine [3]. It subtracts the capacity-weighted mean, sets the analytically zero decay rate and its fluctuation coefficient to zero, and applies

$$\boldsymbol{T}(t) = \bar{T}\mathbf{1} + M^{-1/2} V e^{-t\Lambda} V^T M^{1/2}(\boldsymbol{T}_0 - \bar{T}\mathbf{1}). \tag{5}$$

This removes time truncation error but retains eigensolver roundoff and spatial error. The dense eigenvectors require $O(N^2)$ storage; no performance or large-scale solver claim is made. A separate C++ wrapper would add little here: the expensive linear algebra already runs in compiled libraries.

3 Reproducible experiment

The left/right lengths are 0.06/0.03 m, conductivities 16/2 W/(m K), capacities $4 \times 10^6/2 \times 10^6$ J/(m^3 K), and initial temperatures 500/300 K. Set $R = 2 \times 10^{-4}$ m^{2}K/W, giving $e_L = 8000$, $e_R = 2000$ J/(m^{2}K$\sqrt{\text{s}}$) and $t_c = 0.1024$ s. Each slab has $n = 20, 40, 80, 160, 320$ uniform cells. We record 101 logarithmically spaced times from 10^{-3} to 10 s plus a separate one-second evaluation. Heat is the right slab's capacity-weighted temperature increase. Relative errors use the half-space values in Eqs. (2)–(3).

Domain truncation is assessed at 10 s by doubling both lengths and cell counts from 160 to 320 per slab, preserving cell widths. A separate homogeneous unit-domain test evolves exact cell averages of $\cos(\pi x)$ to $t = 0.1$ with $k = c = 1$ and compares with $e^{-\pi^2 t}\cos(\pi x)$ cell averages on $10, 20, 40, 80$ cells. This smooth test does not assume uniform convergence at the discontinuous initial contact.

4 Measured results

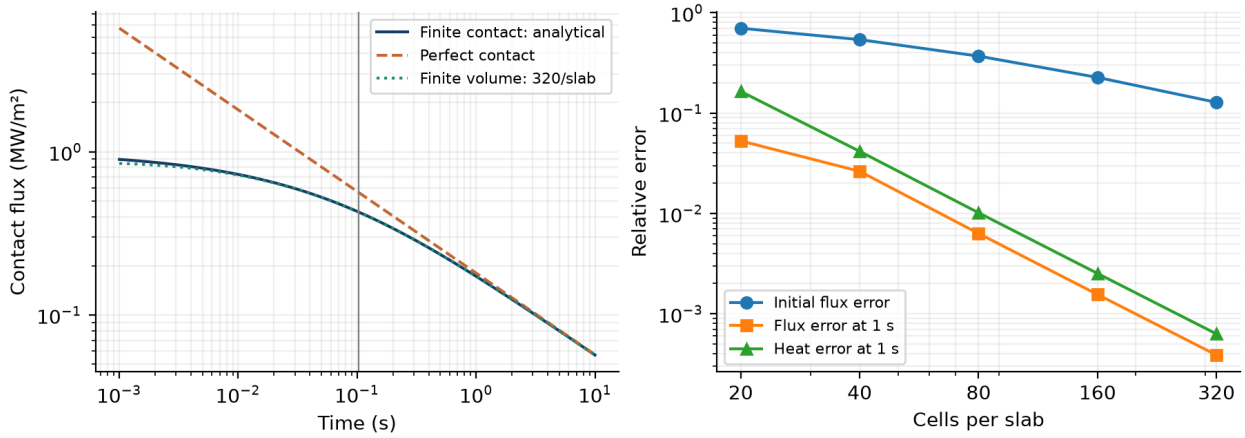


Figure 1: Left: finite-contact reference and finest finite-volume flux, compared with perfect contact; the vertical line marks t_c. Right: error at the initial instant and at one second. All plotted data are generated by the project.

Cells/slab	$q_h(0)/q(0)$	$E_q(1)$	$E_Q(1)$
20	0.29907	5.275e-02	1.643e-01
40	0.46043	2.627e-02	4.152e-02
80	0.63054	6.302e-03	1.014e-02
160	0.77341	1.554e-03	2.520e-03
320	0.87223	3.872e-04	6.290e-04

Table 1: Initial flux ratio and relative one-second errors. $E_q = |q_h/q - 1|$ and $E_Q = |Q_h/Q - 1|$. Scientific notation is used for errors.

The reference values at one second are $q = 1.7244 \times 10^5$ W/m^2 and $Q = 2.7634 \times 10^5$ J/m^2. Perfect contact predicts 30.666% more transferred heat at this time. This difference is a model-assumption effect, not an error of the finite-contact discretization. The finest flux refinement order at one second is 2.0049; the independent smooth cell-average test gives order 1.9998.

The initial finite-volume flux is exactly determined by the series resistance:

$$\frac{q_h(0)}{q(0)} = \frac{1}{1+\eta_h}, \qquad \eta_h = \frac{h_L/(2k_L) + h_R/(2k_R)}{R}. \tag{6}$$

Thus the coarse mesh resolves less than one third of the initial flux, and even the finest mesh remains below nine tenths (Table 1). The finest observed positive-time flux error over the recorded interval is about 5.46%; its heat error reaches 7.54%. The corresponding one-second errors are much smaller. A single late observation is consequently insufficient to certify early contact resolution.

On the recorded local run, the small-chain modal/dense-exponential discrepancy is 2.8422×10^{-13} K. The maximum energy-balance residual over all contact histories is 0.001 821 9 J/m^2 (roundoff relative to large sensible energies). Domain doubling changes the final flux by a relative 7.0024×10^{-13}; this is below a useful roundoff-level distinction and is not an estimate accurate to all displayed digits. These diagnostics are stored in machine-readable form and tested against explicit cross-platform budgets.

5 Discussion and limitations

Equation (6) identifies a practical spatial criterion for measuring the initial contact law: η_h must be small. Requiring an initial relative flux deficit below a chosen ϵ implies $\eta_h < \epsilon/(1-\epsilon)$. This is an initial-state condition, not a universal bound on every later temperature or heat-transfer error. The declining contact flux and growing diffusion layers explain why later agreement can look better than the initial ratio. The refinement table demonstrates this effect for one deliberately specified pair of materials.

Three error levels remain separate. First, the two-cell solution, dense exponential and discrete heat/energy identities verify the finite-dimensional solver. The constant mode is treated analytically, although roundoff leakage from other computed modes remains. Second, the half-space comparison measures spatial error together with any finite-domain mismatch; the domain-doubling check bounds that concern only empirically at the reported final time and spacing. The homogeneous cosine experiment independently verifies smooth continuum convergence. Third, none of these calculations validates a real contact resistance or the assumptions in Eq. (1).

The model excludes roughness evolution, pressure dependence, thermal expansion, temperature-dependent properties, latent heat and interfacial heat capacity. The two synthetic bodies have no asserted material identity. Equal cell counts and the chosen lengths happen to match the diffusive length ratio, but the method accepts arbitrary positive cell properties. Extreme floating-point scales, nearly disconnected chains and very large grids are outside the tested range. There is no inference of R from noisy data and no measured runtime speedup.

6 Conclusion and reproducibility

A late-time flux check can hide an unresolved thermal contact. For this study, the one-second flux approaches second-order spatial convergence while the initial deficit follows a first-order half-cell resistance effect. Reporting both the physical crossover t_c and the mesh ratio η_h makes the distinction explicit. Further work could test temperature-dependent contact laws or compare with independently measured interface data.

Install with `python -m pip install -e ".[dev]"`; run `contactscale -output results` to regenerate all study data and the figure. Run `python -m pytest` for the test suite and execute `scripts/check_results.py` with Python for numerical reproduction. Build this paper by executing `scripts/build_paper.py`; the script accepts pdfLaTeX or Tectonic, rejects undefined references and overfull boxes, enforces at most five pages and records source/PDF hashes. CI runs the software on Windows and Linux and compiles the paper independently. No authorship, affiliation or peer review is asserted by this manuscript.

References

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